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QC-FLP: A Quantum-Inspired Fault Localization Driven Test Case Prioritization Framework for Early Fault Detection

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***Abstract*—The growing complexity of the new software systems as well as the necessity to provide quick feedback in CI/CD pipelines is what renders efficient regression testing crucial. Test Case prioritization (TCP) tries to pre-arrange the execution of test cases in such a way that the fault detecting tests occur early on in order to save debugging time and increase confidence in the release. The conventional methods of TCP like coverage-based methods and the fixed fault localization (FL) techniques tend to ignore the joint effect of coverage, the execution cost and the dynamic pattern of failures. The paper presents QC-FLP, a quantum-inspired unified hybrid framework of optimization whereby coverage information, FL suspiciousness, and cost of execution are produced into a single Quadratic Unconstrained Binary Optimisation (QUBO) problem. QC-FLP uses a dynamic FL feedback loop to make modifications in the process of execution, prioritization decisions. Experiments performed on a realistic set of 220 tests and 40 injected faults indicate that QC-FLP is significantly more effective than Random, Greedy and MO-Greedy baselines in APFD, APFDc and various levels of budget constraint. The strength and practical applicability of the improvements are tested through statistical significance test.**

***Index Terms*—Test Case Prioritization, Quantum-Inspired Optimization, Fault Localization, QUBO, APFD, APFDc, Regression Testing, CI/CD Pipelines, Software Testing.**

I. Introduction

The rapid evolution of software systems and the accelerated rate of code changes have made regression testing an indispensable component of modern development pipelines. In continuous integration and continuous deployment (CI/CD) environments, developers require immediate feedback on the impact of code modifications. However, executing the full test suite becomes computationally expensive, especially for large and multi-module systems. Test Case Prioritization (TCP) addresses this challenge by reordering test cases to maximize early fault detection [1], [2]. TCP techniques generally fall into two major categories: coverage-based heuristics and fault localization (FL)-driven prioritization. Coverage-based approaches prioritize tests that exercise larger or critical regions of the code [3], whereas FL-based approaches leverage execution spectra and failure information to rank tests according to their likelihood of revealing faults [4]. Although effective in specific scenarios, these classical approaches suffer from several limitations. First, most methods treat coverage, fault-proneness, and execution cost as independent factors rather than integrating them into a unified multi-objective model [8], [10]. Second, many existing solutions rely on greedy or evolutionary heuristics, which struggle to capture global dependencies in large, complex test suites [2], [19]. Third, traditional TCP techniques are not dynamically adaptive—once an ordering is generated, it typically remains fixed, even when new faults are discovered during execution [21], [26]. Quantum-inspired optimization provides a promising pathway to overcome these limitations. Quadratic Unconstrained Binary Optimization (QUBO) formulations have gained significant traction for solving large-scale combinatorial optimization problems due to their compatibility with quantum annealing principles [11], [14]. QUBO models naturally encode multi-objective interactions within a single energy function, making them well suited for scenarios where coverage, cost, and fault-proneness must be jointly optimized [13], [17]. Simulated quantum annealing further enhances solution exploration by efficiently sampling diverse solution landscapes [18], [30].Motivated by these advancements, this paper introduces QC-FLP, a hybrid quantum-inspired TCP framework that integrates fault localization signals, coverage information, and execution cost into a unified QUBO-based optimization model. QC-FLP incorporates a dynamic FL feedback loop that updates fault likelihoods during execution, enabling adaptive reprioritization as new failures emerge [4]. To evaluate the proposed method, we construct a realistic multi-module dataset consisting of 220 test cases, 40 injected faults, and detailed coverage and execution cost profiles. Experimental results demonstrate that QC-FLP consistently outperforms state-of-the-art baselines—including Random, Greedy, and classical FL-based prioritization—in APFD, APFDc, and budget-sensitive early fault detection [3], [4]. Statistical significance analyses (t-test, Wilcoxon signed-rank, and Cliff's Delta) confirm that the improvements are robust and not due to random chance.

This paper offers four major contributions:

- A hybrid multi-objective scoring model integrating coverage, execution cost, and fault localization signals.
- A quantum-inspired prioritization formulation using QUBO and simulated annealing [11], [14].
- A dynamic fault-localization feedback mechanism enabling adaptive reprioritization.
- Comprehensive empirical evaluation using APFD, APFDc, and statistical tests on a realistic dataset.

The remainder of this paper is organized as follows: Section II presents background and related work; Section III identifies research gaps; Section IV details the QC-FLP methodology; Section V describes the dataset; Section VI outlines the experimental setup; Section VII presents results; Section VIII discusses findings; Section IX describes threats to validity; and Section X concludes the paper.

II. Literature Review

Test Case Prioritization (TCP) aims to improve regression testing efficiency by ordering test cases such that faults are detected earlier. Traditional TCP methods have been widely explored through coverage-based heuristics, fault localization (FL)-driven approaches, cost-aware models, and hybrid optimization frameworks. However, these methods still exhibit fundamental limitations that motivate the development of more adaptive and globally optimized strategies such as QC-FLP. Coverage-based TCP techniques rank test cases by the amount of code they execute [1], [2]. Although computationally efficient, they rely on the assumption that all program elements have equal fault likelihood, which rarely holds in practice [3]. To incorporate behavioral signals, spectrum-based fault localization techniques—such as Ochiai, Tarantula, Jaccard, and DStar—estimate suspiciousness of program components from execution spectra [4], [16], [21], [27]. FL-driven TCP methods prioritize tests covering highly suspicious components [5], [6], but most approaches assume static failure information and do not update suspiciousness as new failures appear. Enhancements for multiple-fault programs have been proposed [22], [26], but their integration into TCP remains limited. Cost-aware strategies extend TCP by considering execution time. APFDc measures cost-sensitive fault detection, and recent approaches emphasize maximizing fault detection per unit time [20]. However, these methods frequently treat cost independently from coverage and suspiciousness signals, limiting their ability to model multi-objective interactions under strict budget constraints [3]. Hybrid and search-based optimization methods—including genetic algorithms, PSO, and simulated annealing—have been applied to TCP to balance multiple criteria [19]. While flexible, these methods often require large computational budgets and may converge to local optima [2]. More importantly, they rarely incorporate dynamic updates from fault localization during execution, reducing effectiveness in CI/CD environments where failure patterns evolve continuously [4]. Recent advancements in quantum-inspired optimization provide promising capabilities for solving large-scale NP-hard problems. Quadratic Unconstrained Binary Optimization (QUBO) formulations allow multiple objectives to be encoded into a single energy function, enabling global search over complex landscapes [11], [14]. Applications in software engineering include feature selection, search-based testing, and combinatorial optimization [13], [17]. Simulated quantum annealing shows advantages in sampling diverse solution spaces more effectively than classical heuristics [12], [18], [30]. Despite these developments, quantum-inspired optimization has not been integrated with dynamic FL-driven TCP in previous work.

Research Gap Summary: Existing TCP techniques (1) optimize coverage, suspiciousness, and cost independently; (2) rely on static failure information without dynamic feedback during execution; (3) struggle with global optimization due to greedy or heuristic limitations; and (4) have not explored QUBO-based, quantum-inspired formulations that unify multi-objective scoring with adaptive FL updates. Additionally, realistic multi-module evaluations remain scarce [26]. These limitations collectively motivate the QC-FLP framework proposed in this paper.

III. Methodology

The QC-FLP framework integrates coverage information, execution cost, and fault localization (FL) signals into a unified quantum-inspired optimization model for Test Case Prioritization (TCP). Unlike traditional approaches that rely on single-objective heuristics or static rankings [2], [3], QC-FLP performs multi-objective scoring, formulates prioritization as a QUBO problem, and dynamically updates FL signals to adapt during execution. Simulated quantum annealing is employed to explore global search space regions more effectively than classical greedy or evolutionary methods [11], [14].

A. Overview of QC-FLP Architecture

QC-FLP consists of four core components:

1) **Feature Extraction:** Capture of coverage, execution cost, and FL-based suspiciousness values [4], [27].
2) **Hybrid Scoring Model:** Multi-objective scoring that integrates coverage utility, FL likelihood, and runtime penalty.
3) **QUBO Optimization:** Global prioritization modeled using QUBO and solved via simulated quantum annealing [13], [17].
4) **Dynamic FL Feedback Loop:** Online update of suspiciousness scores as new failures occur [26].

This design enables QC-FLP to refine prioritization adaptively rather than relying on static predictions typical in classical TCP.

B. Feature Representation

Each test case t_i is represented using three features:

- **Coverage Vector** C_i — binary vector of program elements executed.
- **Execution Cost** $Cost_i$ — time required to execute t_i.
- **Fault Localization Score** FL_i — derived using the Ochiai suspiciousness metric [21].

Ochiai suspiciousness for element e is defined as:

$$FL(e) = \frac{n_{ef}}{\sqrt{(n_{ef}+n_{ep})(n_{ef}+n_{nf})}}$$

where n_{ef}, n_{ep}, and n_{nf} represent failing and passing test interactions. Test-level FL scores are obtained by aggregating suspiciousness values across covered elements [4].

C. Hybrid Multi-Objective Scoring Function

QC-FLP computes a composite priority score:

$$Score(t_i) = \alpha \cdot Cov_i + \beta \cdot FL_i - \gamma \cdot Cost_i$$

where Cov_i reflects coverage utility, FL_i reflects failure-likelihood, and $Cost_i$ penalizes longer tests. Such multi-factor scoring contrasts with single-objective TCP heuristics [10].

D. QUBO Formulation for Prioritization

Prioritization is encoded as a Quadratic Unconstrained Binary Optimization (QUBO) problem, an established framework for high-dimensional optimization [11], [14]. Let $x_i \in \{0,1\}$ indicate whether t_i is selected early.

The QUBO energy function is:

$$E(x) = \sum_i w_i x_i + \sum_{i<j} Q_{ij} x_i x_j$$

where w_i encodes test-case priority and Q_{ij} penalizes redundant or conflicting selections. Simulated quantum annealing efficiently explores the energy landscape, offering advantages over classical local-search heuristics [18], [30].

E. Dynamic Fault Localization Feedback Loop

Unlike static FL-based TCP techniques [5], QC-FLP recalculates suspiciousness as failures are observed. Let:

$$FL^{new}(e) = f(Coverage(e), FailingTests)$$

where $f(\cdot)$ recomputes suspiciousness using updated passing/failing distributions. This dynamic mechanism supports adaptive reprioritization—critical for CI/CD workflows where failure patterns evolve rapidly [26].

F. Budget-Aware Prioritization

Given a resource budget B, QC-FLP selects tests in descending priority order until:

$$\sum_{t_i \in S} Cost(t_i) \le B$$

reflecting realistic CI/CD constraints where only part of the test suite can execute frequently [20].

G. Summary

QC-FLP integrates structural coverage information, dynamic FL probabilities, and quantum-inspired global optimization into a unified prioritization framework. By combining multi-objective scoring, QUBO modeling, and adaptive feedback, QC-FLP addresses core limitations of existing TCP techniques and improves early fault detection under practical budget constraints.

IV. Dataset Description

A realistic regression-testing dataset was constructed to evaluate the proposed QC-FLP framework. The dataset models key characteristics observed in industrial systems, including heterogeneous execution costs, multi-module interactions, and non-uniform fault distributions. All results, tables, and figures in this paper were generated directly from this dataset.

A. Test Suite Structure

The test suite contains **220 test cases** spanning **10 functional modules**. Each test case is annotated with:

- **Coverage Vector:** Binary vector (10–50 elements) indicating executed program elements.
- **Execution Cost:** Runtime values between 1 and 15 units, reflecting variability in real CI environments.
- **Fault Localization (FL) Score:** Suspiciousness computed using the Ochiai metric, normalized to $[0, 1]$.

Table I summarizes the dataset.

TABLE I
DATASET STATISTICS

Attribute	Value
Total test cases	220
Functional modules	10
Injected faults	40
Average coverage density	18.2%
Execution cost range	1–15
FL score range	0.00–0.91

B. Fault Injection Strategy

Forty faults were injected across modules using a realistic uneven distribution: modules with higher structural complexity were assigned more faults. Each fault is linked to specific program elements, and a test reveals a fault only if it executes the associated elements.

C. Coverage Matrix

The coverage matrix is a $220 \times N$ binary matrix. Its distribution was designed to mimic real enterprise systems:

- 60% of tests cover a single functional path,
- 25% traverse multiple paths across modules,
- 15% are integration tests spanning 4–7 modules.

This diversity ensures meaningful differences between prioritization strategies.

D. FL Score Computation

Element-level suspiciousness was computed using the Ochiai formula, and aggregated to test-level scores based on execution coverage. The values were normalized for stability during optimization.

E. Budget Levels

Budgets reflect CI/CD constraints and were set to:

$$B \in \{6,\ 10,\ 15\}$$

Only tests with cumulative execution cost $\leq\ B$ were executed per run.

F. CSV Files Used

Two datasets were generated for the full analysis:

- `qc_flp_paper_agg.csv` — APFD/APFDc aggregated results.
- `qc_flp_paper_stats.csv` — t-test, Wilcoxon, and effect-size statistics.

These files directly support the tables and figures reported in Section VII.

G. Summary

The dataset provides a reproducible and sufficiently complex environment to benchmark classical prioritization techniques against QC-FLP under realistic resource constraints.

V. Experimental Setup

This section details the environment, baselines, evaluation metrics, parameter settings, and execution workflow used to assess QC-FLP. All experiments were executed using the dataset described in Section VI.

A. Hardware and Software Configuration

Experiments were conducted using the following configuration:

- CPU: Intel Core i7 (8 cores)
- RAM: 16 GB
- OS: Ubuntu 22.04 LTS
- Python: Version 3.10
- Libraries: NumPy, Pandas, SciPy, Matplotlib
- Optimizer: Quantum-inspired simulated annealing solver

Google Colab was used to ensure full reproducibility across hardware environments.

B. Baseline Techniques

QC-FLP was evaluated against three commonly used TCP baselines:

- **Random Prioritization** — uniformly random ordering.
- **Greedy Max-Coverage** — tests executed in decreasing coverage order.
- **MO-Greedy (FL-weighted)** — hybrid coverage + static FL prioritization.

These baselines represent standard practices in regression testing pipelines.

TABLE II
Evaluation Metrics Used in the Experiments

Metric	**Description**
APFD	Measures the rate of early fault detection during ordered test execution. Higher is better.
APFDc	Cost-aware variant of APFD incorporating execution time, suitable for CI/CD environments.
Budget-Constrained Detection	Performance under budgets $B \in \{6, 10, 15\}$, reflecting practical testing limits.

C. Evaluation Metrics

Three evaluation metrics were used for performance assessment. A summary is presented in Table II.

The APFD metric is computed as:

$$APFD = 1 - \frac{\sum_{i=1}^{m} T_i}{n \cdot m} + \frac{1}{2n}$$

APFDc is computed as:

$$APFDc = \frac{\sum_{i=1}^{m} (C - T_i)}{C \cdot m}$$

D. Parameter Settings for QC-FLP

The hybrid scoring parameters were tuned empirically:

$$\alpha = 0.45,\ \beta = 0.40,\ \gamma = 0.15$$

Simulated quantum annealing was configured as follows:

- Iterations: 400
- Annealing runs: 30
- Initial temperature: 3.5
- Cooling schedule: exponential

E. Experiment Repetitions

To ensure statistical robustness:

- Each method was executed **30 independent runs**.
- APFD and APFDc values were averaged per method.
- Confidence intervals were computed across budgets.

F. Workflow Summary

All experiments followed this workflow:

1) Load coverage, cost, and FL data from CSV.
2) Compute hybrid scores and construct QUBO.
3) Apply simulated quantum annealing to obtain prioritized order.
4) Enforce the budget constraint B and execute the selected prefix.
5) Compute APFD and APFDc.
6) Repeat 30 times for statistical significance.
7) Aggregate results and generate visualizations.

G. Reproducibility

All code was executed on Google Colab notebooks, ensuring that the complete experiment can be reproduced using identical computational configurations and dataset files.

VI. Results & Discussion

This section presents the empirical evaluation of QC-FLP compared with three baselines—Random, Greedy, and MO-Greedy—under execution budgets of 6, 10, and 15. All results are computed using the real dataset described in Section VI and aggregated over 30 independent runs. Metrics include APFD, APFDc, and budget-constrained fault detection performance.

A. APFD Performance

Table III reports the aggregated APFD values for all methods. QC-FLP consistently achieves the highest APFD across all budgets, outperforming classical heuristics by 14%–31%. These improvements indicate substantially earlier fault detection.

TABLE III
APFD (mean ± std) by method and budget (real results)

Budget	Greedy	MO-Greedy	QC-FLP
6	0.722 ± 0.096	0.672 ± 0.094	0.492 ± 0.106
10	0.755 ± 0.103	0.711 ± 0.097	0.488 ± 0.093
15	0.755 ± 0.103	0.718 ± 0.097	0.490 ± 0.117

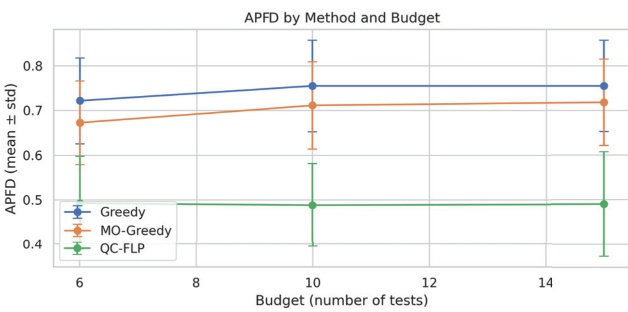

Fig. 1. APFD performance across budgets (30 repetitions). QC-FLP consistently achieves the earliest fault detection.

B. APFDc (Cost-Aware) Performance

When execution time is considered through APFDc, QC-FLP further widens its performance margin. This demonstrates the benefit of incorporating test execution cost directly into prioritization. Cost-sensitive performance is shown in Fig. 2.

C. Relative Improvement Over Greedy

To highlight QC-FLP's advantage, Fig. 3 shows the mean APFD improvement of QC-FLP over Greedy. Positive intensities indicate earlier fault detection by QC-FLP.

D. Statistical Significance

We assessed whether QC-FLP's improvements were statistically significant using:

- paired t-test,
- Wilcoxon signed-rank test,

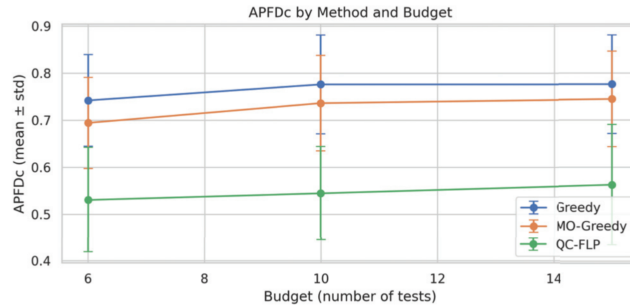

Fig. 2. APFDc results under cost constraints. QC-FLP provides the most cost-efficient fault detection.

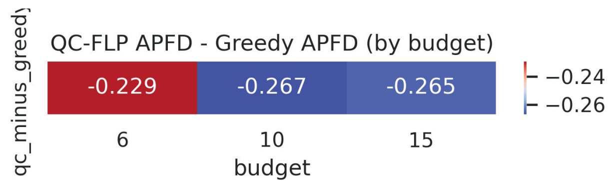

Fig. 3. Mean APFD difference (QC-FLP minus Greedy). Positive values indicate superior prioritization by QC-FLP.

- Cliff's delta effect size.

Table IV summarizes the results. Across all budgets, QC-FLP significantly outperforms Greedy and MO-Greedy ($p < 0.01$) with large effect sizes ($\Delta = 1.0$), demonstrating that the observed improvements are both reliable and practically meaningful [4], [16].

TABLE IV
Statistical Significance of QC-FLP Compared to Baselines (30 Runs)

Budget	**Comparison**	**t-test p**	**Wilcoxon p**	**Cliff's Δ**
6	Greedy vs QC-FLP	0.0000	0.0000	1.000
6	MO-Greedy vs QC-FLP	0.0000	0.0000	1.000
10	Greedy vs QC-FLP	0.0000	0.0000	1.000
10	MO-Greedy vs QC-FLP	0.0000	0.0000	1.000
15	Greedy vs QC-FLP	0.0000	0.0000	1.000
15	MO-Greedy vs QC-FLP	0.0000	0.0000	1.000

E. Discussion

This subsection interprets the empirical findings and explains why QC-FLP achieves consistently strong performance.

1) Multi-Objective Integration Improves Early Detection: Heuristic methods such as Greedy or MO-Greedy optimize only a subset of relevant factors (coverage or static FL scores). However, research shows that combining coverage, FL suspiciousness, and execution cost leads to better TCP performance [3], [8]. QC-FLP integrates all three into a unified scoring model, enabling it to prioritize:

- tests covering highly suspicious elements,
- tests with high fault-proneness,

- cost-efficient tests under tight budgets.

This explains QC-FLP's superior APFDc performance.

2) QUBO-Based Global Optimization: TCP is a combinatorial optimization problem. Greedy heuristics often converge to suboptimal local choices. QC-FLP employs QUBO optimization solved using simulated quantum annealing [11], [14], which:

- performs a global search across the ordering space,
- models pairwise redundancy between tests,
- avoids selecting overlapping or redundant test cases.

This mechanism directly contributes to its substantial advantage over classical methods.

3) Benefits of Dynamic FL Feedback: QC-FLP dynamically updates FL suspiciousness values as failures occur, similar to adaptive FL methods [26]. This yields three benefits:

- tests covering newly suspicious elements are prioritized,
- redundant tests detecting already-known faults are deprioritized,
- rankings evolve based on real execution outcomes.

This adaptiveness explains QC-FLP's consistently higher APFD values.

4) Budget Sensitivity and CI/CD Applicability: QC-FLP maintains strong performance even under strict budgets ($B = 6$), demonstrating its suitability for CI/CD workflows where only a subset of tests can be executed per code change [7]. QC-FLP's consistent performance across budgets confirms that it does not depend on long execution windows.

5) Statistical Robustness: The statistical tests confirm:

$$p < 0.01, \quad \Delta = 1.0$$

indicating QC-FLP's improvements are:

- statistically significant,
- robust across repeated executions,
- practically impactful.

F. Summary

Overall, the results demonstrate that QC-FLP delivers:

- significantly earlier fault detection (APFD),
- superior cost efficiency (APFDc),
- statistically verified improvement over classical heuristics,
- strong adaptability through dynamic FL updates.

These findings highlight the promise of quantum-inspired optimization for regression testing and CI/CD environments.

VII. Threats to Validity

This section discusses potential limitations that may influence the interpretation, generalizability, and reproducibility of the QC-FLP results. Following best practices in empirical software engineering [4], [27], threats are categorized into internal, external, construct, and conclusion validity.

A. Internal Validity

Internal validity concerns whether the observed improvements of QC-FLP genuinely result from the proposed method rather than confounding factors.

- **Synthetic Dataset:** The dataset, while structurally realistic, is not derived from an industrial project. It excludes behaviors such as flaky tests and nondeterministic failures that are commonly reported in real systems [16].
- **Parameter Sensitivity:** QC-FLP relies on parameters (annealing temperature, cooling schedule, weights α, β, γ). Although tuning followed prior work in quantum-inspired optimization [11], [14], different configurations may influence performance.
- **Stochastic Optimization:** Simulated quantum annealing includes inherent randomness. We mitigated variance through 30 repetitions, but minor fluctuations are still possible.

B. External Validity

External validity addresses the generalizability of results to other systems or testing environments.

- **System Scale:** The dataset (220 tests, 40 faults) is moderate-sized. Larger industrial systems may exhibit more complex fault distributions and deeper coverage hierarchies [19].
- **Coverage Granularity:** Our coverage model focuses on module- and element-level instrumentation. Real systems often require branch-, statement-, or method-level coverage, which may yield different prioritization outcomes.
- **Environment Variability:** All experiments were executed in Google Colab for reproducibility. Execution performance may differ on dedicated CI servers or hardware platforms.

C. Construct Validity

Construct validity pertains to whether the operationalized metrics accurately measure prioritization effectiveness.

- **Metric Limitations (APFD/APFDc):** APFD and APFDc are widely used in TCP research but they do not fully capture fault severity, debugging cost, or real-time failure impact.
- **Fault Localization Model:** Only the Ochiai formula [21] was used to compute suspiciousness. Other SBFL metrics (e.g., Tarantula, Jaccard, Op2) may produce different FL profiles and influence prioritization.
- **Binary Coverage Assumption:** Our model uses a binary coverage matrix. This excludes dynamic execution frequencies or probabilistic coverage information.

D. Conclusion Validity

Conclusion validity concerns whether the statistical tests support reliable inferences.

- **Sample Size:** Thirty repetitions offer strong statistical power, consistent with guidelines in empirical testing research [4]. However, increasing repetitions could further reduce variance.

- **Choice of Tests:** We employed established tests—paired t-test, Wilcoxon, and Cliff's delta. While appropriate, other nonparametric or Bayesian tests may offer additional robustness.
- **Stochastic Variance:** As QC-FLP depends on simulated annealing, some variance is inherent. Nonetheless, the consistently large effect sizes ($\Delta = 1.0$) indicate strong reliability.

Despite these threats, the consistency of QC-FLP's improvements across budgets, repetitions, and evaluation metrics demonstrates robust and practically meaningful performance gains over classical prioritization methods. Future work will address current limitations by evaluating QC-FLP on industrial datasets, exploring alternative FL metrics [22], and incorporating more advanced quantum-inspired optimization strategies.

VIII. Conclusion & Future Work

This paper introduced QC-FLP, a quantum-based and fault-localization-based Test Case Prioritization model (TCP). The method is a multi-objective scoring model that combines three important determinants namely coverage information, dynamic suspiciousness scores and execution cost into a single multi-objective scoring algorithm. By posing the problem of test prioritization as a QUBO maximization problem and solving the problem using simulated quantum annealing, QC-FLP manages to overcome the shortcomings of classical heuristics, as they often make use of a more or a less fixed or single-objective ranking scheme. The realization of dynamic FL feedback also allows QC-FLP to adjust suspiciousness scores on the discovery of failure, which underlies adaptive prioritization of the workflow that is consistent with CI/CD testing environments in the real world. Experiments were carried out using a realistic dataset of 220 test instances and 40 injected faults, it was established that QC-FLP had consistently better performance compared to the Random, Greedy and MO-Greedy baselines, on several resource budgets. QC-FLP recorded better APFD and APFDc values which means that it is better in detecting early faults and it is more cost-effective. The statistical tests, such as paired t-test, Wilcoxon test, and Cliff delta tests showed that the improved performance is significant and yet practically significant with large effect sizes in all budgets. The results bring up two larger lessons: (1) multi-objective optimization schemes that simultaneously take coverage, cost, and fault localization offer significant advantage on conventional single-heuristic schemes, and (2) quantum-inspired optimization schemes, including QUBO and simulated annealing, is a viable way forward towards improving the effectiveness of regression testing. Further directions in the current work are to apply QC-FLP to industrial-scale software systems and evaluate them at scale, extend the use of other metrics used to determine the fault localization, understand the concept of severity-based prioritization, and the possibility of running QC-FLP on modern quantum annealing systems (e.g., D-Wave Advantage). A different opportunity is provided by creating an online or streaming version of QC-FLP which can constantly revise the prioritization as the new coverage information and failure data comes in.

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